

ELOOP TOKEN (ELP)

Digital Infrastructure for Reverse Logistics Compliance & Monetization

Institutional & Technical Whitepaper — 2026 Edition

Reverse Logistics • Blockchain Validation • Compliance Infrastructure

Confidential — Institutional Distribution

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1. Executive Summary

Eloop is a digital compliance infrastructure for reverse logistics and electronic waste (e-waste) traceability. It converts physical disposal events into legally defensible records, verifiable environmental data, and programmable digital units (ELP).

Eloop is not a speculative ESG token. It is a compliance infrastructure backed by a utility token.

Core Value Proposition

- Regulatory compliance automation for companies subject to Brazil's PNRS (Law 12.305/2010) and global EPR mandates
- Immutable chain-of-custody: QR scan → certified weight → blockchain hash → audit-ready report
- Token utility: ELP is issued exclusively after validated physical events — no speculative minting
- Enterprise-facing SaaS model: companies pay for compliance infrastructure, not for cryptocurrency

2. Market Context & Structural Gap

2.1 Regulatory Growth

Global e-waste generation reached 62 million tonnes in 2022, with less than 22% formally recycled (Global E-Waste Monitor, 2024). Regulatory pressure is accelerating:

- Extended Producer Responsibility (EPR) mandates expanding across 78+ countries
- Brazil: PNRS (Law 12.305/2010) + Decreto 10.936/2022 + Acordos Setoriais de Logística Reversa
- EU: WEEE Directive 2012/19/EU, CSRD digital reporting obligations
- Canada: EPRA provincial mandates + ESG disclosure requirements

2.2 Structural Gap

Companies subject to reverse logistics obligations currently manage compliance with spreadsheets, PDFs, and unverifiable declarations. This creates:

- No immutable chain-of-custody for disposed equipment
- Audit risk: data can be altered, lost, or fabricated
- No standardized digital reporting compatible with SINIR (Brazil) or international ESG standards
- Inability to monetize verified environmental data for ESG reporting

Eloop closes this gap by converting physical disposal events into cryptographically verifiable compliance records — transforming regulatory obligation into auditable digital infrastructure.

2.3 Market Sizing (Brazil First)

Segment	Volume	Eloop Addressable
Companies under PNRS obligation	~12,000 large companies	SAM: ~2,000 initial targets
Licensed recyclers / cooperatives	~1,800 certified operators	SOM: ~200 pilot network
E-waste volume (Brazil/year)	~2.2 million tonnes/year	Pilot target: 500–1,000 t

3. System Architecture — Four Layers

Eloop operates across four integrated layers, each mitigating a specific risk vector in the compliance chain:

Layer 1 — Physical Validation

Captures the real-world disposal event with certified data:

- QR Code: unique lot ID generated exclusively by the Eloop system (single-use, tamper-evident)
- INMETRO-certified scale: weight captured via direct API, IoT gateway, or digitally signed device — manual entry not permitted
- Equipment category: classified per ABNT NBR 16156
- Photographic evidence: mandatory scale readout + lot image
- Operator ID: licensed recycler or cooperative with valid environmental permit

Layer 2 — Data Capture & IoT Integration

Converts physical events into structured, validated digital records:

- API-first architecture: Node.js backend + REST/Webhook integration
- IoT gateway (modular): Raspberry Pi / ESP32 for direct scale integration (Levels 2–3)
- Level 1 fallback: Glide mobile app (offline-capable) for low-connectivity cooperatives
- Data stored: PostgreSQL (primary) + Firebase (mobile sync)

Layer 3 — Blockchain Anchoring

Creates cryptographic proof of each validated event:

- Network: Polygon (low cost, EVM-compatible, <\$0.01 per transaction)
- Process: event metadata → SHA-256 hash → stored in IPFS → hash anchored in Polygon smart contract
- Blockchain stores the fingerprint, not the file — privacy-preserving by design
- Infrastructure: Thirdweb or Alchemy SDK — blockchain remains invisible to end users

Layer 4 — Enterprise Compliance Interface

Delivers audit-ready output to enterprise clients:

- Compliance dashboard: tonnes tracked, validations, CO2e avoided, ESG metrics
- Exportable reports: PDF + CSV, SINIR-compatible format
- Public verification link: each event verifiable on Polygonscan
- Enterprise API: B2B integration with ERP systems

The client never manages a crypto wallet. Blockchain is operational infrastructure — the compliance report is what they buy.

4. Formal Chain-of-Custody Model & ELP Distribution Equations

4.1 Validated Event Structure

Each physical disposal event is formally represented as:

$$e_i = (w_i, c_i, v_i, h_i, t_i, p_i)$$

Where:

- w_i = certified weight (kg), INMETRO-scale
- c_i = equipment category (per ABNT NBR 16156)
- v_i = validator ID (licensed recycler / cooperative)
- h_i = SHA-256 hash anchored on Polygon
- t_i = Unix timestamp of validation
- p_i = photographic evidence hash (IPFS CID)

A validation is accepted ($V = 1$) only when all fields are present and pass integrity checks:

$$V(e_i) = 1 \quad \text{iff} \quad w_i > 0 \wedge c_i \in C \wedge v_i \in \text{Validators} \wedge h_i \neq \emptyset \wedge t_i \text{ valid}$$

4.2 ELP Issuance Model — Weight-Based Distribution

ELP tokens are issued strictly after a validated event. The baseline issuance formula is:

$$\text{ELP_issued}(e_i) = w_i \times \gamma(c_i) \times \alpha \times \beta(v_i)$$

Where:

- w_i = certified weight in kg
- $\gamma(c_i)$ = gravimetric complexity factor by category (see Table 4.2a)
- α = base rate (ELP per kg) — governance-controlled parameter
- $\beta(v_i)$ = validator quality multiplier (0.8 – 1.2), based on audit score

Table 4.2a — Gravimetric Complexity Factor $\gamma(c)$ by Equipment Category

Equipment Category	$\gamma(c)$ Factor	Basis	Example Devices
Smartphones & tablets	2.5	High rare-metal density	iPhone, Samsung Galaxy
Laptops & notebooks	2.0	High component complexity	MacBook, ThinkPad
Desktop computers & servers	1.8	High volume + mixed materials	Tower PCs, rack servers
Monitors & displays	1.4	Glass + PCB	LCD/LED monitors
Large appliances (white goods)	1.0	Baseline — bulk metal	Refrigerators, washing machines
Cables & peripherals	0.8	Low rare-metal content	USB cables, keyboards
Batteries (separated)	3.0	Hazardous + high recovery value	Li-ion, lead-acid
Printers & copiers	1.5	Toner + PCB	HP, Canon, Xerox

$\gamma(c)$ values are based on gravimetric composition studies (ABRELPE, WEEE Forum) and may be updated by governance vote. Batteries are tracked separately due to hazardous material protocols.

4.3 Base Rate α — Governance-Controlled Parameter

The base rate α (ELP per kg) is a governance parameter adjusted to maintain tokenomic balance:

$$\alpha = \text{ELP_budget_period} / \Sigma(w_i \times \gamma(c_i)) \text{ for all validated events in period}$$

- Initial reference value: $\alpha = 5.0$ ELP/kg (pilot phase, subject to governance)
- Example: 1 smartphone (0.2 kg) $\rightarrow 0.2 \times 2.5 \times 5.0 \times \beta = 2.5$ ELP (at $\beta=1.0$)
- Example: 100 kg of mixed large appliances $\rightarrow 100 \times 1.0 \times 5.0 \times \beta = 500$ ELP
- α is reviewed quarterly by the Eloop governance council; changes require 72h notice

4.4 Validator Quality Multiplier $\beta(v)$

$\beta(v)$ incentivizes validator compliance quality:

$$\beta(v_i) = 0.8 + (0.4 \times \text{Qscore}(v_i) / 100)$$

- Qscore: 0–100, based on uptime (40%), zero-discrepancy rate (40%), audit pass rate (20%)
- β range: 0.80 (minimum, underperforming) $\rightarrow$ 1.20 (maximum, exemplary)
- Validators below $\beta = 0.85$ for 3 consecutive months trigger governance review

4.5 Triple Validation Requirement

ELP issuance is blocked unless all three conditions are met simultaneously:

$$\text{ISSUE}(e_i) = 1 \text{ iff } \text{QR_valid}(e_i) = 1 \wedge \text{Weight_certified}(e_i) = 1 \wedge \text{Hash_recorded}(h_i) = 1$$

Failure in any single condition blocks token release for that event. Partial events are logged but not tokenized.

4.6 Aggregate Supply Control

Total ELP in circulation at any time T is bounded by the Compliance Incentive Pool (25% of 1B = 250M ELP):

$$\Sigma \text{ELP_issued}(t) \leq 250,000,000 \text{ for all } t \leq T$$

Additional supply from other pools (Enterprise, Investors, Founders) follows separate vesting and governance schedules, never event-based issuance.

5. Tokenomics — ELP Supply Architecture

Total Fixed Supply: 1,000,000,000 ELP (hard cap — no inflation mechanism)

Allocation	%	Strategic Role	Application Logic
Compliance Incentive Pool	25%	Event-backed issuance	ELP issued only after validated reverse logistics events (QR + certified weight + hash). Prevents speculative minting. Physical-to-digital bridge.
Enterprise Utility & Liquidity	25%	Corporate acquisition	Companies acquire ELP to register volumes, generate audit reports, and access compliance modules. Creates real corporate demand.
Investors / Institutional	20%	Capital formation	Structured allocation for strategic long-term investors. Includes lock-up and vesting. Funds infrastructure development.
Founders & Team	15%	Governance	Vesting schedule: 25% at 3 months, 25% at 6 months, 50% at 12 months. Conditional on roadmap milestone delivery.
Infrastructure & Impact Fund	15%	Validator network & R&D	Funds physical validation layer expansion, API development, carbon metrics integration, and ecosystem growth.

Vesting Schedule — Founders & Team (15%)

- 25% released at month 3 (operational milestones confirmed)
- 25% released at month 6 (MVP in production with pilot data)
- 50% released at month 12 (enterprise subscription active)

Team Allocation Breakdown (15% = 150,000,000 ELP)

- Founder / Lead: 50M ELP (5%) — strategy, representation, institutional
- Co-Founder (Legal & Contracts): 20M ELP (2%)
- Co-Founder (Operations): 20M ELP (2%)
- Technical Partner 1 — Blockchain/MVP: 10M ELP (1%)
- Technical Partner 2 — ESG Marketing: 10M ELP (1%)
- Technical Partner 3 — Governance & Standards: 10M ELP (1%)
- Technical Partner 4 — Product Quality: 10M ELP (1%)
- Reserve (future technical hires): 20M ELP (2%)

6. Fraud Mitigation Architecture

Eloop's security model operates across three independent layers. Fraud would require compromising all three simultaneously — a cost that exceeds any regulatory penalty avoided:

Layer	Mechanism	Protection
Physical	INMETRO-certified scale + licensed facility	Weight cannot be fabricated without compromising certified hardware
Automation	IoT + API data capture + SHA-256 hashing	Data alteration detectable — any change invalidates hash
Cryptographic	Blockchain anchoring + IPFS storage	Immutable timestamp — retroactive modification impossible

7. KPI Framework — Technical, ESG & Business

7.1 Platform & Usage KPIs

KPI	Unit	Target (Year 1 Pilot)
Active Users (MAU)	count	50–200 (cooperatives + operators)
Validated Events	events/month	>500 by month 6
Scan-to-Validation Rate	%	>92%
Average Validation Time	minutes	<60 min end-to-end
System Availability	%	99% uptime SLA
Cost per Transaction (Polygon)	USD	<\$0.01 per hash

7.2 Environmental KPIs

KPI	Unit	Target (Year 1 Pilot)
E-waste Tracked	kg / tonnes	500–1,000 t in pilot
Devices Processed	units	>10,000 devices
Recovery Rate	%	>85% formal recycling
CO2e Avoided	kg CO2e	Calculated per WARM / ISO 14064
CO2e per kg Processed	kg CO2e / kg	~1.5–3.5 kg CO2e / kg e-waste
Annual Impact Growth	%	>40% YoY

7.3 Business & Token KPIs

KPI	Unit	Target
Licensed Partners Onboarded	count	2–3 recyclers + 1 cooperative (pilot)
Enterprise ESG Reports Generated	reports/month	>10 by month 6
Revenue per Enterprise Partner	USD/month	\$500–2,000 (subscription)
ELP Issued (on-chain)	ELP/month	Proportional to validated tonnes
Traceability Coverage	%	>95% of events with full hash
Regulatory Alignment Status	status	PNRS compliant + SINIR-ready

8. Technology Stack — MVP Roadmap

8.1 MVP Architecture (8 Weeks)

Week	Module	Tool / Stack	Output
1–2	Cooperative mobile app	Glide (no-code) + Google Sheets	QR scan + scale photo + weight entry → structured data
3–4	Backend validation	Make.com / n8n automation + Thirdweb SDK	Auto hash generation + Polygon registration + ELP simulation
5–6	Enterprise dashboard	Lovable.dev (AI-generated React) + Supabase	Compliance metrics, ESG KPIs, PDF report export
7–8	Pilot with 1 cooperative	On-site deployment + real data	500+ validated events, first real compliance report

8.2 Full Production Stack

- Backend: Node.js + FastAPI (microservices)
- Database: PostgreSQL (primary) + Firebase (mobile sync)
- Blockchain: Polygon + Thirdweb SDK + IPFS (Pinata or Web3.Storage)
- IoT: Raspberry Pi / ESP32 + MQTT broker (Mosquitto) for Level 2–3 validators
- Frontend: React (Lovable.dev scaffold) + Supabase auth
- Reports: Metabase or Power BI embedded + PDF auto-generation
- Smart Contract: ERC-20 ELP token (verified on Polygonscan: 0x6b01aDa61be146d090cdA56Be7B73EF557f5DE26)
- QR system: single-use QR per lot, generated server-side, bound to event metadata

8.3 CO2e Calculation Methodology

Carbon avoidance per validated event is calculated using EPA WARM model factors:

$$\text{CO2e_avoided}(e_i) = w_i \times \gamma(c_i) \times \text{EF}(c_i)$$

Where $\text{EF}(c_i)$ is the emission factor (kg CO2e / kg) per category (reference: EPA WARM v15, ISO 14064-2):

- Mixed electronics: ~2.5–3.5 kg CO2e / kg avoided vs. landfill
- Batteries (Li-ion): ~4.0–6.0 kg CO2e / kg (hazardous + recovery value)
- Large appliances: ~1.2–2.0 kg CO2e / kg

CO2e calculations are provided as estimated environmental impact metrics for ESG reporting. Independent third-party verification is recommended for formal carbon credit claims.

9. Deployment Roadmap — Brazil First

Phase 1 — Controlled Pilot (Months 1–6)

- 2 licensed recyclers + 1 structured cooperative onboarded
- Level 1 MVP deployed: Glide app + automated backend + Polygon anchoring
- Target: 500–1,000 tonnes validated
- First enterprise compliance reports generated
- Governance parameters (α , γ) calibrated with real data

Phase 2 — Regional Expansion (Months 7–18)

- Dashboard integration with SINIR reporting format
- Enterprise subscription model launched (B2B SaaS)
- Level 2 validators (semi-automated scales) onboarded
- Carbon metrics layer integrated (CO₂e reporting)
- InovAtiva Brasil / Artemisia accelerator applications

Phase 3 — National Scale (Months 19–36)

- Multi-validator network (50+ licensed operators)
- Governance token voting for α and γ parameters
- International expansion groundwork: Canada (EPRA), European WEEE
- ELP listing on decentralized exchanges (Uniswap liquidity pool active)

10. Regulatory Framework & Compliance Alignment

Regulation	Jurisdiction	Eloop Alignment
Lei 12.305/2010 (PNRS)	Brazil	Full — mandatory reverse logistics compliance layer
Decreto 10.936/2022	Brazil	SINIR-compatible reporting format
ABNT NBR 16156	Brazil	Equipment classification standard used in $\gamma(c)$ table
INMETRO scale certification	Brazil	Required for all physical validation events
ISO 14001	International	Validator eligibility criterion
ISO 14064-2	International	CO ₂ e calculation methodology
WEEE Directive 2012/19/EU	EU	Alignment for future European expansion
EPRA Provincial Mandates	Canada	Target market for Phase 3 expansion

11. Intellectual Property & Protection

- Blockchain timestamp: SHA-256 hash registered on Bitcoin blockchain via OpenTimestamps
- License: CC BY-NC-ND 4.0 — attribution required, no commercial use without permission, no derivatives
- Smart contract address (Polygon):
0x6b01aDa61be146d090cdA56Be7B73EF557f5DE26
- Creator wallet: 0x5E9E4562A18d6eB80A7f80EaA8b2DBa2a8fDF97F
- GitHub repository: github.com/Maffioletti24/eloop-token-licenciamento
- Trademark registration (INPI Brazil): pending — in process
- International trademark (WIPO Madrid Protocol): planned post INPI grant
- NDA required for all technical partners and developers

12. Team & Contact

Name	Role	Focus
Sérgio Maffioletti	Founder & Project Lead	Sustainability, environmental blockchain, solid waste management
Leandro Lacerda	Co-Founder	Contracts, legal, regulatory compliance

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